PROJECT 9 · PHASE 1

Indoor Location Prediction Using Wi-Fi Signals

Team T084 | Section D

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Where is this phone indoors?

- GPS works poorly indoors: walls and roofs weaken satellite signals
- Wi-Fi is in most buildings; a phone hears many WAPs (access points)
- One scan = the signal strength (RSSI, in dBm) of every WAP
- Building and floor: classification; (x, y) position: regression

Wi-Fi scan

RSSI of 520 WAPs, most of them “not heard”

→

Model

kNN, Random Forest or XGBoost

→

Answer

building · floor · (x, y) position

Data: UJIIndoorLoc (UCI): 3 buildings, 19,937 training scans, 1,111 test scans

What others did, and the gap

Work	What they did
RADAR (2000)	Match a new scan to the closest stored fingerprints
UJIIndoorLoc (2014, 2015)	Public multi-building dataset; kNN with many distances
Classifier comparison (2015)	kNN best; bagged or boosted trees close behind
Neural networks (2017–2019)	Autoencoder, one shared network, CNN on UJIIndoorLoc

Gap 1 · Random splits look too good

One network: 98.2% building-and-floor accuracy on a random split, 91.1% on the later test file

Gap 2 · Our tasks are learned separately

Our three task models share nothing; Kim et al. (2018) trained one shared network on UJIIndoorLoc

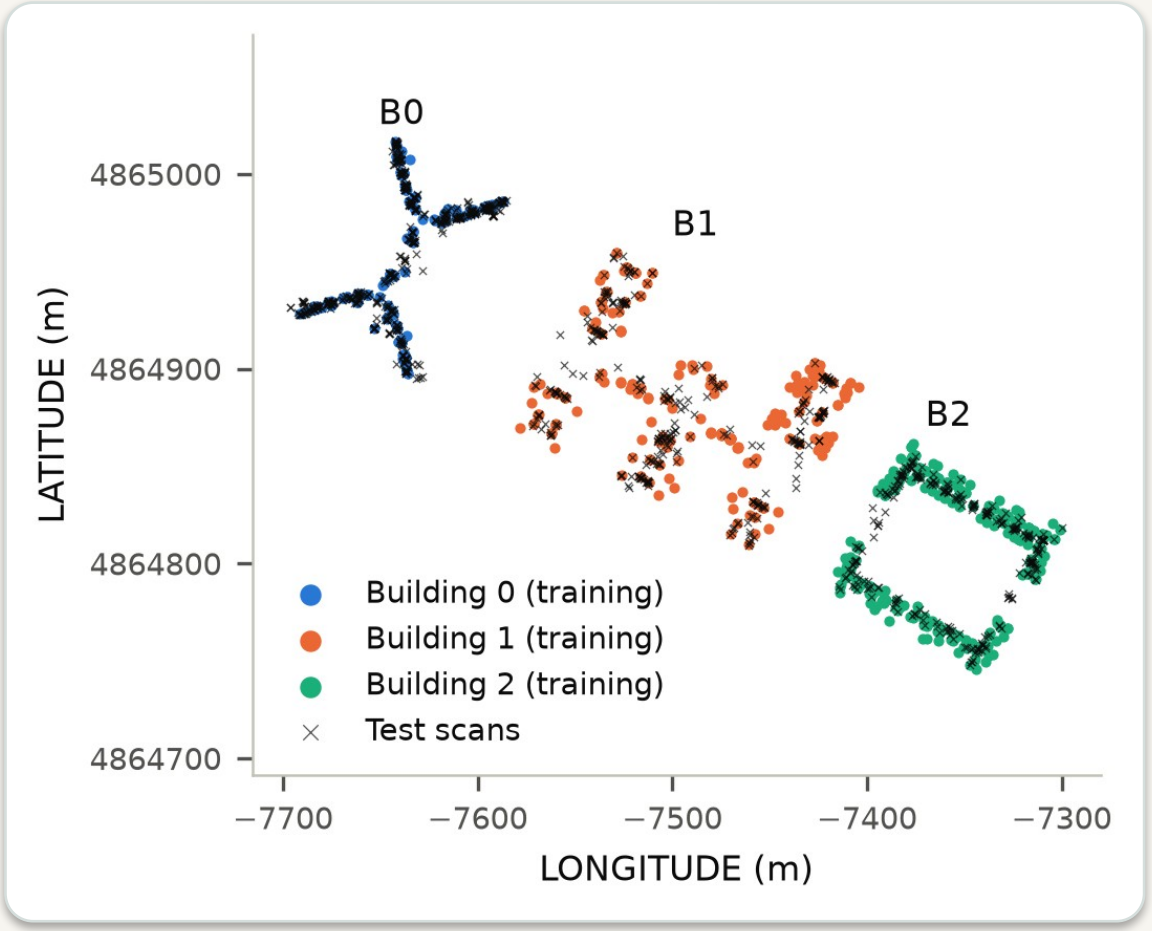
Sources: Bahl & Padmanabhan 2000; Torres-Sospedra et al. 2014, 2015; Bozkurt et al. 2015; Nowicki & Wietrzykowski 2017; Kim et al. 2018; Song et al. 2019

The data: UJIIndoorLoc

	Training	Test set
Scans	18,719 (cleaned)	1,111
Phones	16	11 (9 new)
Recorded	May–Jun 2013	Sep–Oct 2013

- 3 buildings, 13 building-floor classes
- 520 WAPs; a scan hears 17.8 on average
- 96.6% of training readings: “not heard”
- 64.3% of test scans from new phones

Test set = the UCI file validationData.csv
Web-Mercator meters: 1 m ≈ 0.77 m on the ground



Cleaning and a fair split

1 · Cleaning (training file only)

Rule	Removed
No WAP heard	76
Exact duplicate	634
Reading of −30 dBm or stronger	508

19,937 → 18,719 scans

- The test set is never cleaned
- Kept: the 464 WAPs heard in training

2 · A fair split: user-group cross-validation (CV)

- 5 folds; each holds out whole users with their phones
- Shared phone = one group: 18 users → 16 groups

Baseline RF	Building	Floor	Mean 2D error
Random CV	100.0%	99.9%	3.67 m
User-group CV	**99.7%**	**78.0%**	**15.87 m**
Test set	99.8%	91.2%	12.52 m

Random CV promises **3.67 m**; the test set gives **12.52 m**

Features: from a scan to location clues

RSSI · 464 columns

Signal strength of each WAP heard in training; not heard = −110 dBm

Signal statistics · 3 columns

WAPs heard, strongest reading, mean of the heard readings

AP-affinity (our idea) · 8 columns

1 · Learn where each WAP is

From the training scans that hear it: building, floor, location

2 · Average the profiles

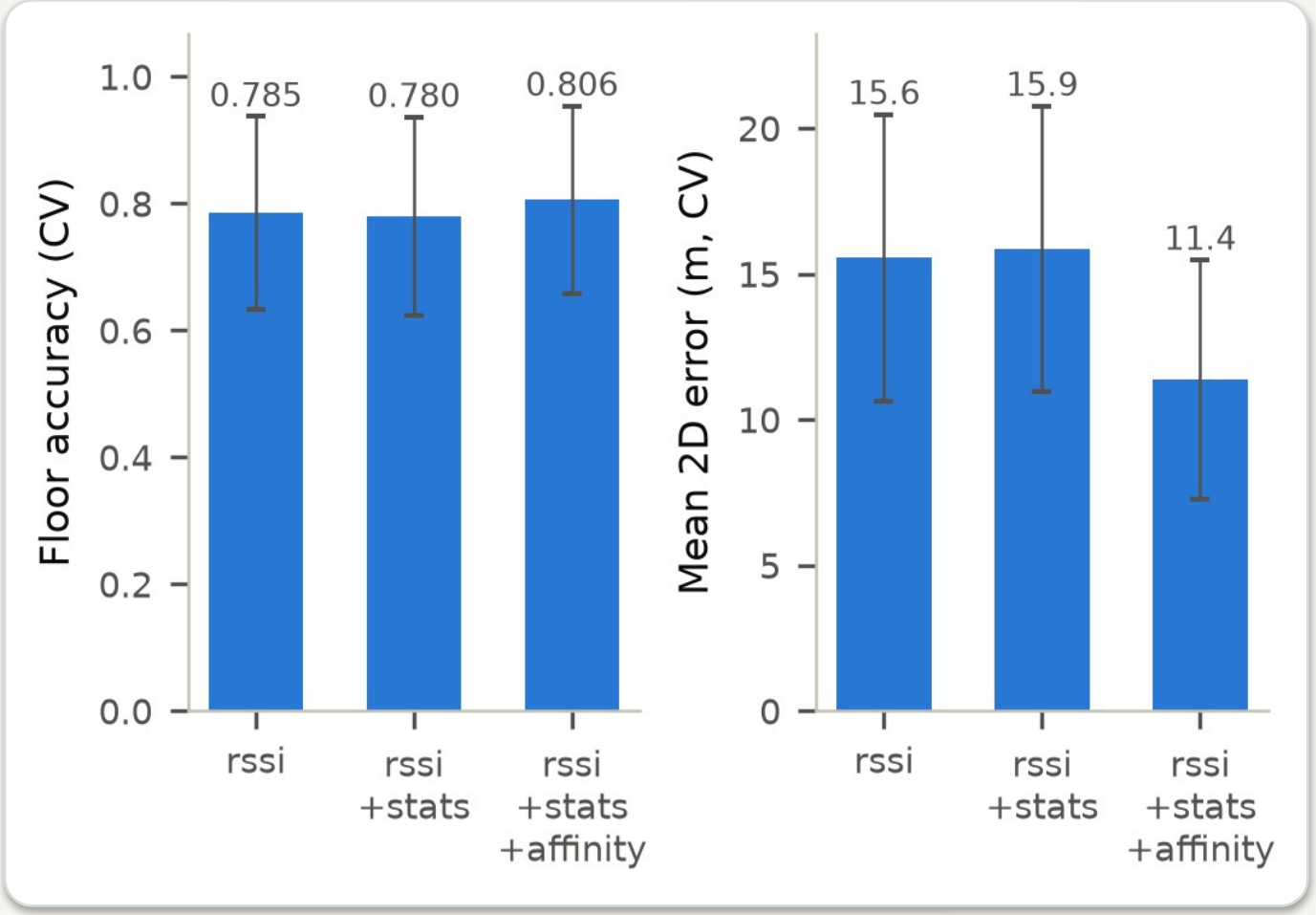
Of the WAPs the scan hears; stronger readings count more

3 · Add the strongest WAP

Learned location of the strongest known WAP the scan hears

AP-affinity is learned from training scans only, again inside every CV fold, so held-out scans never shape their own features.

Feature check: AP-affinity helps



Baseline RF settings · user-group CV

Adding AP-affinity

15.87 → 11.38 m

CV mean 2D error, 28.3% lower; floor accuracy 78.0% → 80.6%

Adding signal statistics

15.55 → 15.87 m

No gain: CV mean 2D error slightly higher; floor accuracy 78.5% → 78.0%

Four models, tuned fairly

Model	Features	Tuned with user-group CV	Settings tried
Baseline RF	rssi + stats	nothing: 300 trees, default settings	1
kNN	rssi	k, weights, distance	28
Random Forest	rssi + stats + affinity	features per split, leaf size	6
XGBoost	rssi + stats + affinity	tree depth, number of trees	6

- Three separate models per method: building, floor and position
- Small grids; each task keeps its own best CV setting

Final-model rule, fixed before testing: lowest CV mean 2D error

Random Forest 11.09 m · XGBoost 11.25 m · kNN 12.75 m → final model: **Random Forest**

Test set results (1,111 scans, used once)

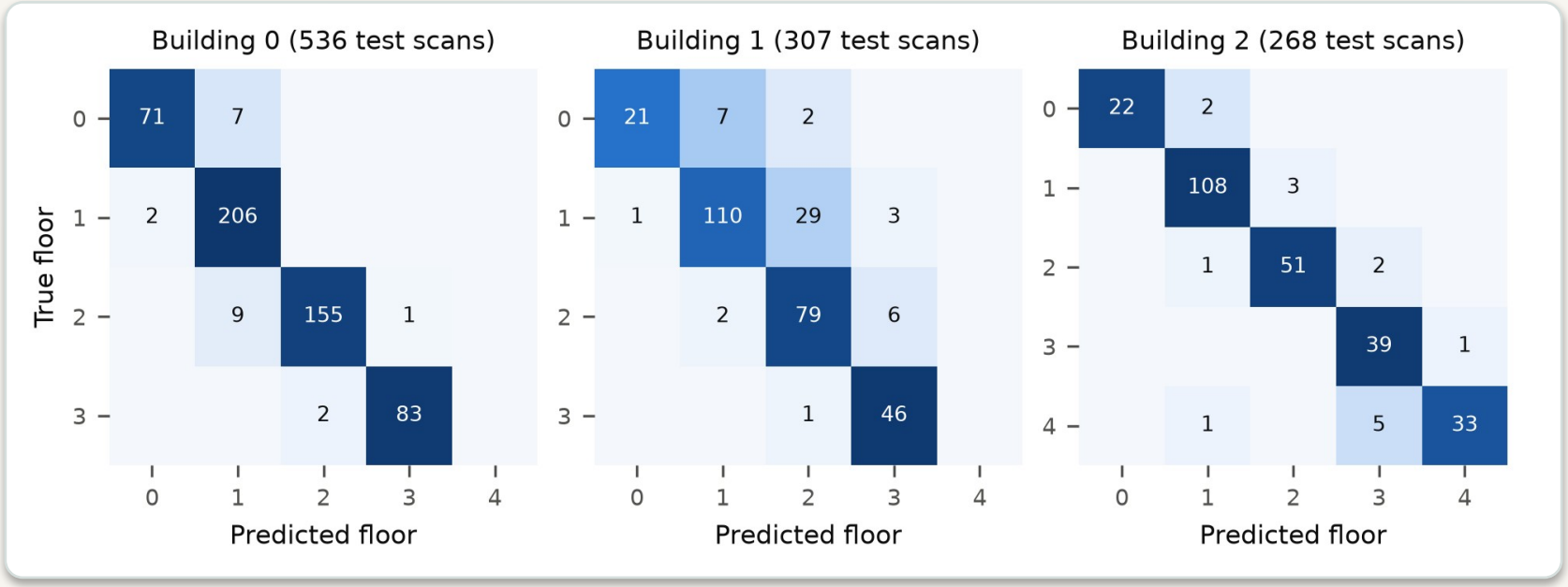
Model	Building	Floor	Mean 2D error	Median 2D error
Baseline RF	99.8%	91.2%	12.52 m	7.70 m
kNN	99.8%	87.5%	9.22 m	5.88 m
Random Forest (final model)	**100.0%**	**92.2%**	**7.42 m**	**5.34 m**
XGBoost	100.0%	95.2%	7.86 m	6.19 m

Random Forest, picked before testing, has the lowest error: **7.42 m** (Baseline RF 12.52 m).
XGBoost gets the most floors right: **95.2%**.

Mean 2D error: straight-line distance between predicted and true position

Where the errors are

Case (Random Forest, test set)	Share of scans	Mean 2D error	Share of error
Building wrong	0.0%	–	0.0%
Building right, floor wrong	7.8%	13.06 m	13.8%
Both right	**92.2%**	**6.94 m**	**86.2%**



81 of 87

floor mistakes are one floor off

8.30 m

on new phones; 5.84 m on seen phones

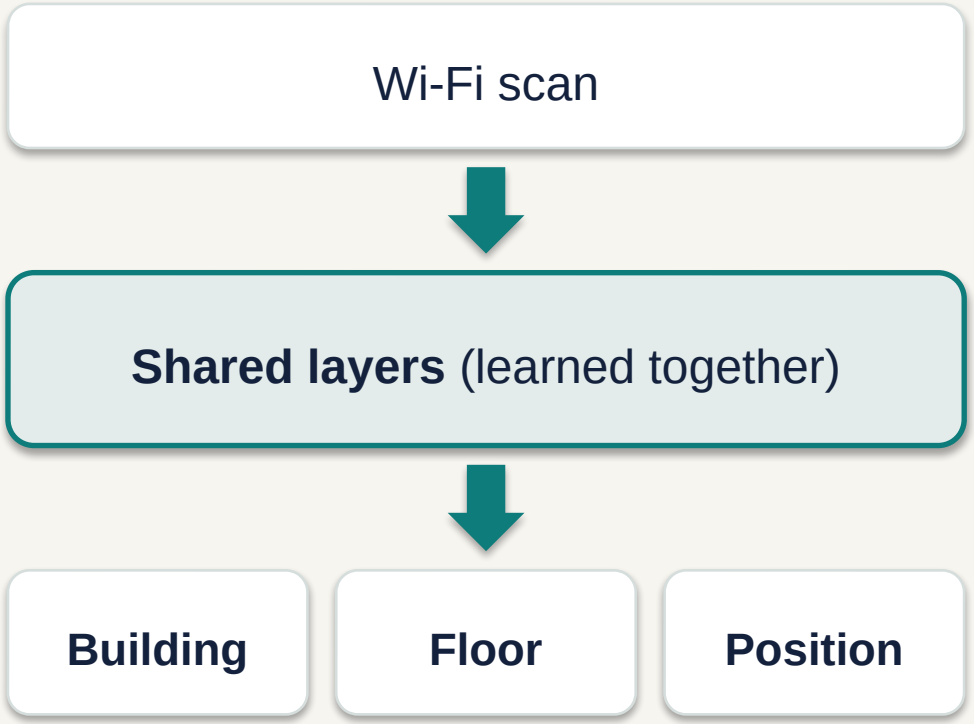
What Phase 1 showed, and Phase 2

What Phase 1 showed

- AP-affinity helped most: CV error 15.87 → 11.38 m
- No single best model: Random Forest for position, XGBoost for floors
- 86.2% of the error is on scans with building and floor right

The gap: our three task models share nothing. AP-affinity shares location knowledge between them, but only by hand.

Thank you!



Phase 2: one multi-task neural network, with the same split, test set and metrics, compared with Phase 1